Spectral Query Spatial: Revisiting the Role of Center Pixel in Transformer for Hyperspectral Image Classification

Ning Chen, Leyuan Fang, *Senior Member, IEEE*, Yang Xia, Shaobo Xia, Hui Liu, and Jun Yue, *Member, IEEE*

***Abstract*— Recently, there have been significant advancements in hyperspectral image (HSI) classification methods employing Transformer architectures. However, these methods, while extracting spectral-spatial features, may introduce irrelevant spatial information that interferes with HSI classification. To address this issue, this article proposes a spectral query spatial transformer (SQSFormer) framework. The proposed framework utilizes the center pixel (i.e., pixel to be classified) to adaptively query relevant spatial information from neighboring pixels, thereby preserving spectral features while reducing the introduction of irrelevant spatial information. Specifically, this article introduces a rotation-invariant position embedding (RIPE) module to integrate random central rotation (RR) and center relative position embedding (PE), mitigating the interference of absolute position and orientation information on spatial feature extraction. Moreover, a spectral-spatial center attention (SSCA) module is designed to enable the network to focus on the center pixel by adaptively extracting spatial features from neighboring pixels at multiple scales. The pivotal characteristic of the proposed framework achieves adaptive spectral-spatial information fusion using the Spectral Query Spatial paradigm, reducing the introduction of irrelevant information and effectively improving classification performance. Experimental results on multiple public datasets demonstrate that our framework outperforms previous state-of-the-art methods. For the sake of reproducibility, the source code of SQSFormer will be publicly available at https://github.com/chenning0115/SQSFormer.**

***Index Terms*— Deep neural network, hyperspectral image (HIS) classification.**

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I. Introduction

HYPERSPECTRAL imaging, as an avant-garde technology, propels the acquisition of high-resolution spectral information [1], [2]. This innovative methodology endows valuable insights into the spectral characteristics of diverse entities and materials on the Earth's surface [3], [4]. Specifically, hyperspectral image (HSI) encompasses spatial and spectral reflectance data pertinent to terrestrial entities, with each pixel corresponding to a unique spectral curve [5], [6]. It is noteworthy that hyperspectral technology extends beyond the perceptual scope of the human eye, allowing for the detection of a broader spectral range and enhancing our ability to discern various features on the Earth's surface [7]. Across multifarious domains such as environmental management, agriculture, geology, and ecology, hyperspectral imaging technology has achieved notable success [8]. The primary objective in HSI classification is the meticulous categorization of each pixel into specific land cover classes [9], [10], a pivotal aspect that plays a paramount role in the majority of hyperspectral imaging applications [11].

Due to the high-dimensional spectral characteristics in HSI, various statistical transformation methods have been proposed to convert the spectral vectors from high-dimensional to low-dimensional feature spaces [12], [13]. These spectral transformation methods can be broadly categorized into two types: methods based on orthogonal transformations (e.g., principal component analysis (PCA) [14] and singular value decomposition (SVD) [15]) and methods based on discriminant transformations (e.g., linear discriminant analysis (LDA) [16], independent component analysis (ICA) [17], and minimum noise fraction (MNF) [12]). It is noteworthy that the aforementioned methods are based on linear transformations. However, due to the potential nonlinear structure in hyperspectral data, researchers have proposed several nonlinear transformation methods aimed at better capturing nonlinear features within the data [18]. These methods encompass kernel discriminant analysis (KDA) [19], manifold hypergraph learning (MHL) [20], and kernel sparse representation (KSR) [21]. Nevertheless, considering the spatial homogeneity and heterogeneity of remote sensing image data, relying solely on spectral feature extraction may struggle to fully leverage the comprehensive characteristics of HSIs. Consequently, researchers have introduced some nonlinear transformation methods that jointly consider spatial and spectral information. These methods include extended morphological profile (EMP) [22], [23], 3-D morphological profile (3D-MP) [24], extended attribute profile (EAP) [25], multiple morphological component analysis (MMCA) [26], and directional morphological profile (DMP) [27].

Traditional HSI classification methods typically rely on manually designed spectral-spatial feature extraction, exhibiting limited performance in complex environments [28]. In recent years, deep learning-based HSI classification has become a research hotspot, providing enhanced capabilities for feature learning and representation [10], [29]. To fully exploit the 3-D data features of HSI, a series of spectral-spatial feature extraction methods combining deep neural networks have been proposed. These methods include deep fully convolutional network [30], [31], diffusion model [11], and deep recurrent neural network (RNN) [32]. Among these, methods based on convolutional neural network (CNN) excel in joint learning of spectral and spatial information, effectively capturing local features in the images [33], [34], [35]. Approaches based on RNN [36], [37] and long short-term memory (LSTM) [38], which are sequential models, demonstrate superior ability to handle temporal dependencies within hyperspectral data, thereby contributing to enhanced classification performance [39].

Recently, a novel deep learning architecture, the Transformer model, has been proposed for handling high-dimensional time series data [40], [41]. The distinctive feature of the Transformer model lies in its self-attention mechanism, which allows the model to dynamically establish relationships between different positions in the sequence [42]. This capability enables the Transformer to excel in capturing long-range dependencies within sequences, contributing to a better understanding of the complex structures in high-dimensional time series data. The inspiration for this model stems from its successful applications in natural language processing, and it has now found success in various domains, including image classification [43], [44], object detection [45], [46], and image segmentation [47], [48]. The recent success of the Transformer model in handling high-dimensional time series data offers novel insights for HSI classification [49]. Several Transformer-based HSI classification methods have been proposed, contributing to the enhancement of HSI classification accuracy [50].

In order to fully exploit the spectral-spatial features of HSI, several methods combining Transformer architectures for spectral-spatial feature extraction have been proposed, including HSI transformer (HiT) [51], spatial–spectral 1DSwin (SS1DSwin) [52], spectral–spatial feature tokenization transformer (SSFTT) [53], spatial–spectral transformer with cross-attention (CASST) [54], and light self-Gaussian-attention (LSGA) [55]. HiT encodes representations along the height, width, and spectral dimensions to capture more spectral-spatial information. SS1DSwin characterizes local and hierarchical spectral-spatial relationships along the spatial and spectral dimensions, revealing local and hierarchical spectral-spatial relationships from two distinct perspectives. SSFTT employs Gaussian distribution weighted tokenization to transform shallow spectral-spatial features into tokenized semantic features. CASST adopts a dual-branch structure to separately extract spatial and spectral features, utilizing a cross-attention mechanism to jointly extract spectral-spatial features. LSGA employs mixed spectral-spatial tokenization to replace patch embeddings, preserving the complete relationships of input image patches and obtaining local feature relationships. However, existing Transformer-based spectral-spatial HSI classification methods face several challenges as follows.

1) In conventional computer vision, image classification involves categorizing the entire image. However, HSI classification focuses on classifying the center pixel. Current Transformer-based methods for HSI classification have not adequately distinguished and been specifically designed to address this distinction.
2) The current approach to spatial information position embedding (PE) in HSI introduces absolute spatial orientation information and absolute positional information between neighboring pixels. This may result in the model overfitting to absolute spatial information.
3) The influx of superfluous spatial neighborhood information can introduce unwarranted interference, thereby detrimentally impacting the performance of HSI classification. Navigating the challenge of judiciously acquiring pertinent spatial information adaptively, while concurrently preserving the extraction of spectral information, constitutes a formidable hurdle in this domain.

To address the aforementioned issues, we propose a spectral query spatial transformer (SQSFormer) framework. This framework adaptively extracts relevant spatial information from neighboring pixels based on the center pixel (i.e., the spectral vector of the pixel to be classified), preserving effective spatial features while reducing the introduction of ineffective information. Specifically, SQSFormer consists primarily of a rotation-invariant PE (RIPE) module and a spectral-spatial center attention (SSCA) module. The RIPE module incorporates center relative PE and random central rotation (RR). The former enhances the capture of relevant spatial information by considering the relative positions to the center pixel, while the latter promotes the learning of features at different angles through random rotations around the center pixel. These elements effectively mitigate unnecessary spatial noise associated with absolute PE. The SSCA module comprises multiple layers of center-aware (CA) Transformer Blocks, with each block utilizing the output corresponding to the center pixel label as the spectral-spatial feature. Finally, a Multilevel Center Pixel Embedding module is employed to integrate the spectral-spatial features generated by CA Transformer Blocks across different layers, creating a multiscale spectral-spatial feature. SQSFormer achieves adaptive fusion of spectral-spatial information, reducing the introduction of irrelevant information and effectively enhancing classification performance. The primary contributions of our approach can be summarized in three aspects as follows.

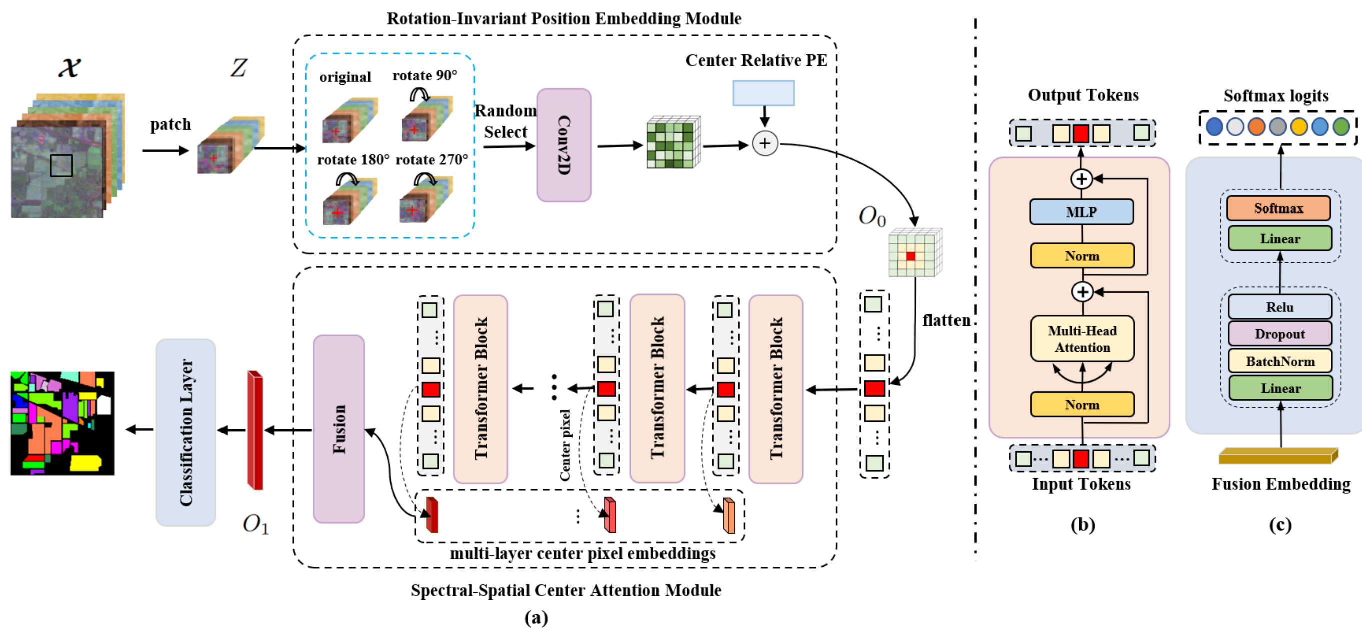

Fig. 1. Framework of SQSFormer. (a) Overall structure of our proposed model. (b) Internal structure of transformer block. (c) Internal structure of classification layer.

1) This article introduces a RIPE Module that integrates data rotation and center relative PE to alleviate the interference of absolute spatial orientation and absolute positional information of neighboring pixels on spatial feature extraction. Specifically, instead of using absolute PE, this article replaces it with center relative PE between neighborhood pixels and the center pixel. Additionally, the input data is rotated to effectively decrease the influence of absolute spatial orientation and absolute spatial positions of neighboring pixels, allowing the network to focus on extracting spatial and spectral features from the center pixel.
2) This article proposes a SSCA Module to enhance the specialized role of the transformer on the center pixel. The module replaces the frequently used class token in the vision transformer (ViT) structure with a center pixel token, enabling the network to focus on the center pixel. Specifically, in each transformer layer, the center pixel embedding representing the spectral feature is used as the Query to adaptively obtain spatial features from neighborhood pixel embeddings through an attention mechanism, effectively integrating spatial and spectral features. This structure can effectively reduce spatial information in neighboring pixels that is irrelevant to classification.
3) This article proposes a novel HSI classification framework called SQSFormer. Compared with previous methods, this framework is capable of adaptively extracting spatial features based on the center pixel and reducing the interference of irrelevant spatial information. Experimental evaluations on multiple public datasets demonstrate that our framework outperforms previous state-of-the-art algorithms in terms of classification performance.

II. Methodology

In this section, we will introduce the proposed SQSFormer for HSI classification, including the details of RIPE module and SSCA module.

A. Overview

The overall framework of our proposed method is illustrated in Fig. 1. Given an HSI, denoted as $\mathcal{H} \in \mathbb{R}^{B\times H\times W}$, where H and W represent the height and width of $\mathcal{H}$, respectively, and B indicates the number of spectral channels. In the preprocessing stage, we perform PCA on the HSI data to obtain $\mathcal{X} \in \mathbb{R}^{C\times H\times W}$, where C represents the dimensionality of the spectra after the processing. Assuming that the entire HSI image consists of N labeled pixels $X_N = \{x_1, x_2, \ldots, x_N\}$, where each pixel corresponds to a one-hot vector class label $Y_N = \{y_1, y_2, \ldots, y_N\}$. In order to effectively utilize the spectral and spatial information available in HSI, our framework generates an image patch Z with a patch size of w around each target pixel (which we call **Center Pixel**) for classification, incorporating the neighborhood information as input. The patch set is defined as $Z_N = \{z_1, z_2, \ldots, z_N\} \in \mathbb{R}^{C\times w\times w}$, where N is the total number of image patches, w is the patch size, and C is the number of spectral bands.

Subsequently, we utilize the image patch Z_N as inputs and pass them through two modules: the **RIPE Module** and the **SSCA Module**. The RIPE module performs data transformation, tokenization, and central relative PE on input data to introduce spatial position information, while concurrently mitigating the interference from absolute orientation information and the absolute position information of neighboring pixels. This module also converts the data into a format suitable for the Transformer network structure and produces O_0. The SSCA module takes the flattened O_0 as input and employs the attention mechanism of the Transformer to adaptively acquire relevant spatial information from neighboring pixels, using the center pixel's spectral information as a query. Next, we merge the output embeddings from the center pixel of each layer of the Transformer to obtain the final module output O_1.

Finally, we believe that the O_1 embeddings contain all the effective spatial and spectral information for the pixels to be classified. We pass this embedding through a classifier to obtain the estimated classification probability P for the corresponding pixel and calculate the classification loss through comparison with the class label. The structure of the classifier is depicted as Fig. 1(c). In this study, we use the CE loss as the selected loss function, as shown in the following equation, where K represents the number of categories, and $p_{ij} \in P$:

$$\text{CEloss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{K}\left(y_{ij}\log\left(p_{ij}\right)\right). \tag{1}$$

B. RIPE Module

This article proposes the RIPE module to alleviate potential interference from absolute spatial orientation information and absolute positional information among neighboring pixels on spatial feature extraction. The module capable of incorporating spatial information in this framework mainly include Conv2D and PE before Transformer input. To minimize the influence of spatial noise, the RIPE module adopts two key designs, namely, RR and center relative PE. The former is employed to alleviate the potential presence of absolute orientation information introduced by Conv2D, while the latter, highlights the role of the center pixel as well as helps mitigate potential interference from absolute spatial information among neighboring pixels. Specifically, for the input Patch data Z_N, it is first rotated by a certain angle around the center pixel as the rotation center. Then, Conv2D is used for tokenization, and the resulting tokens are added to the vectors of center relative PE to generate information for inputting into the SSCA structure. The detailed description is provided below.

1) Random Central Rotation: In the case of limited training samples, the absolute spatial orientation information introduced by the samples may be more pronounced, while the absolute orientation may be weakly correlated with the type of land cover, as it is determined by the flight direction of remote sensing platforms. To mitigate the impact of absolute orientation on model training, we propose randomly rotating the input patch data around the center pixel axis by a certain angle (0°, 90°, 180°, 270°). Fig. 2 illustrates the impact of data rotation on Conv2D. Equations (2) and (3), respectively, express the differences in calculation of the same pixel before and after random center rotation by the same Conv2D kernel, where $v'_{i,j}$ and $v'^{R90}_{i,j}$ represent the calculation results of pixel (i, j) before and after rotation, respectively, and w represents the weight of the Conv2D kernel. Through random rotation, Conv2D can perceive multiple orientation information, alleviate the impact of absolute orientation, and be more compatible with center relative PE. In addition, in the case of limited samples, this method can also increase the diversity of training samples, thereby having a positive effect on classification performance

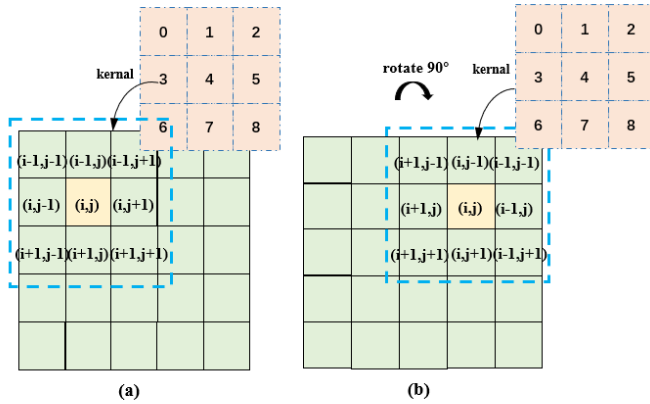

Fig. 2. Differences in calculation of the same pixel before and after random center rotation by the same Conv2D kernel. (a) Before rotation: a 3 × 3 kernel operates on the (i, j) pixel. (b) After 90° rotation: a 3 × 3 kernel operates on the (i, j) pixel corresponding to the pre-rotation state.

$$\begin{aligned} v'_{i,j} &= w_0 v_{i-1,j-1} + w_1 v_{i-1,j} + w_2 v_{i-1,j+1} \\ &\quad + \cdots + w_6 v_{i+1,j-1} + w_7 v_{i+1,j} + w_8 v_{i+1,j+1} \end{aligned} \tag{2}$$

$$\begin{aligned} v'^{R90}_{i,j} &= w_0 v_{i+1,j-1} + w_1 v_{i,j-1} + w_2 v_{i-1,j-1} \\ &\quad + \cdots + w_6 v_{i+1,j+1} + w_7 v_{i,j+1} + w_8 v_{i-1,j+1}. \end{aligned} \tag{3}$$

2) Center Relative PE: Different from traditional image classification tasks in the field of computer vision, HSI classification tasks often consider pixels as the classification units with the objective of determining the land cover category to which each individual pixel belongs. Therefore, in HSI classification tasks, the center pixel plays a crucial role. Generally, the land cover category of the center pixel is strongly correlated with the information of neighboring pixels. As a result, the patch Z_N contains spatial information that is extremely important for the classification task. In the past, researchers attempted to express the spatial positions of individual pixels within the patch using absolute PE, as shown in Fig. 3(a). While this approach accurately characterizes spatial information, it fails to highlight the special position of the center pixel. Additionally, the absolute PE of the neighboring pixels can introduce considerable spatial noise. To address this issue, as depicted in Fig. 3(b), we have designed a center relative PE module for HSI classification tasks, as described in (4), where i and j denote arbitrary pixel column and row indices within input patch, and i_c and j_c represent the column and row indices corresponding to the center pixel. $\mathbb{E}$ denotes the collection of PE, and w represents the patch size

$$\begin{aligned} E_{\text{Pixel}(i,j)} &= E_{\min(|i-i_c|,|j-j_c|)} \\ &\in \mathbb{E} = \{E_k | k = 0, 1, \ldots, (w/2+1)\}. \end{aligned} \tag{4}$$

C. SSCA Module

This article introduces a SSCA Module to enhance the specific role of the transformer on the center pixel. Specifically, for the output O_0 of the RIPE module, we flatten it into individual tokens, where each token still represents each pixel in the original patch space. Subsequently, the data is

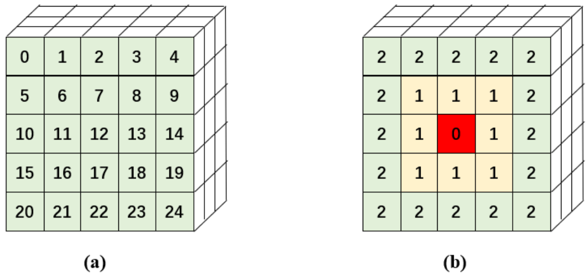

Fig. 3. Different spatial PE methods. (a) Absolute spatial PE. (b) Relative spatial PE.

fed into multiple layers of CA Transformer Blocks. In order to effectively exploit the rich information generated by the different layers of transformers [56], we employ a Fusion structure to integrate the Center Pixel embeddings of each Transformer Block. This fusion process results in a final output vector that contains spectral-spatial fusion features.

1) CA Transformer: As described in Section III-B, unlike in traditional computer vision for image classification tasks, the center pixel plays a crucial role in HSI classification tasks. To highlight the special significance of the center pixel, this article proposes a CA Transformer architecture. The CA Transformer architecture is similar to the ViT, but with a different definition and use of the class token. As shown in Fig. 4, ViT has a dedicated class token that is used as the final classification vector. In traditional visual classification tasks, the classification is performed on the overall image and not on a single pixel, and the class token can better express the extracted features of the entire image. However, when using the class token for HSI classification tasks, the features extracted by the class token often weaken the attention on the center pixel. Therefore, the CA Transformer uses the output corresponding to the center pixel token as the final classification vector.

Another way to understand this architecture is shown in Fig. 5, where we use the center pixel as the Query and the neighboring pixels as the Key/Value for attention calculation. The Query contains spectral information for the center pixel, and the attention mechanism adaptively obtains spatial features from the embedded neighborhood pixels, thereby effectively fusing spatial and spectral features. Specifically, (5) represents the calculation logic of the traditional Transformer structure for Query (Q), Key (K), and Value (V), where d is the corresponding vector dimension. Based on this, in the computation of the CA Transformer, (6) shows the calculation of the Center Pixel vector at the $(l + 1)$ layer. Here, C_{emb}^{l+1} represents the embedding of the Center Pixel in the $l + 1$ layer of the Transformer, which is obtained by using the Center Vector C_{emb}^{l} at the l layer as the Query, and the neighboring pixels within the Patch, K^l, at the l layer as the Key and Value for attention calculation. Finally, the outputs of each layer's Center Pixel are collected and summarized into the C_{emb} set, which serves as the multilevel Spectral-Spatial information representation of the target Pixel

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{5}$$

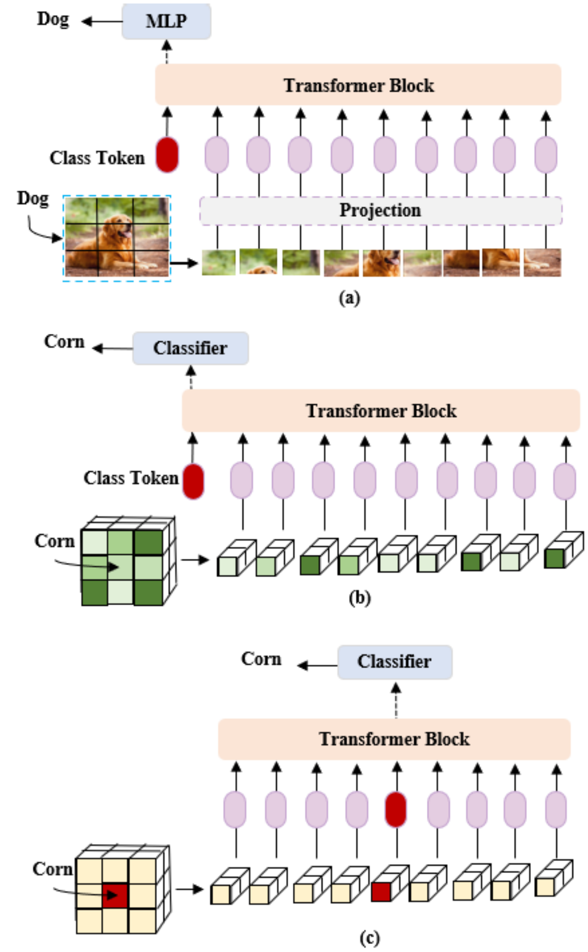

Fig. 4. Comparative illustrative diagram of ViT and CA transformer. (a) ViT incorporates a dedicated class token as the ultimate classification vector. (b) When the ViT structure is applied to HSI classification tasks, it still employs a dedicated class token as the final classification vector. (c) CA transformer employs the output corresponding to the center pixel token as the final classification vector.

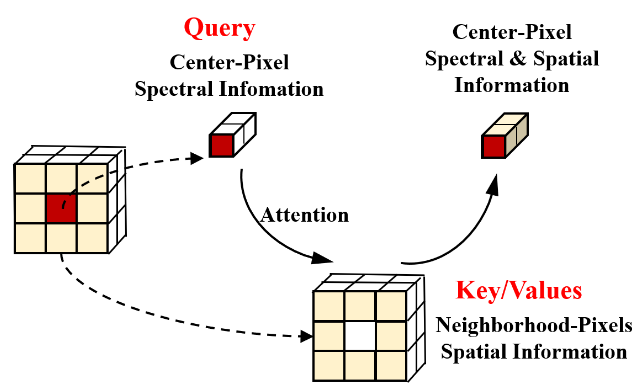

Fig. 5. Use center pixel as query and neighboring pixels as key/value for attention calculation to obtain spectral-spatial information.

$$C_{\text{emb}}^{l+1} = \text{softmax}\left(\frac{C_{\text{emb}}^{l}(K^l)^T}{\sqrt{d}}\right)K^l. \tag{6}$$

2) Fusion: To make better use of the information generated by the CA Transformer Blocks at different layers, we propose to fuse the Center Pixel embeddings of each block. The fusion method is described by (7), where W is an $1 \times n$ matrix and

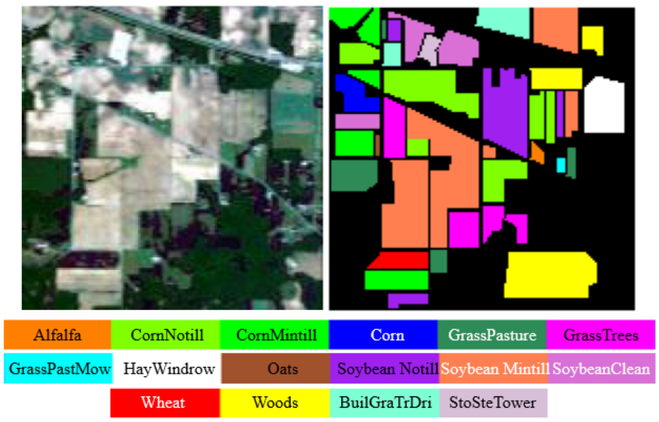

Fig. 6. False-color composite image, the corresponding ground-truth map, and the legend of the IP dataset.

C_{emb} is an $n \times d$ matrix composed of the output center pixel vectors from each layer of the transformer. Here, n represents the number of layers, and d represents the vector dimension

$$O_1 = WC_{\mathrm{emb}}. \tag{7}$$

III. Experiments

A. Experimental Settings

1) Datasets: In order to verify the effectiveness of our algorithm, we applied the algorithm to three public datasets: the Indian Pines (IP) dataset, the Pavia University (PU) dataset, and the WHU-Hi-HongHu (WHHH) dataset.

The dataset known as IP was obtained in 1992 with the utilization of the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Sensor. This dataset covers the northwestern area of Indiana, United States. It consists of 224 spectral bands, spanning a wavelength range of 0.4 to 2.5 μm. The dataset includes a total of 145 $\times$ 145 pixels, with each pixel having a spatial resolution of 20 m. It encompasses 16 distinct land-cover classes. To conduct our analysis, we eliminated 24 bands related to water absorption and noise, thereby selecting a subset of 200 bands. The false color image and the classification groundtruth are shown in Fig. 6.

The PU dataset was acquired during the year 2001 using the reflective optics system imaging spectrometer (ROSIS) Sensor. The dataset encompasses the geographical extent of PU situated in Northern Italy. It comprises a total of 115 spectral bands, with 12 noise bands being excluded from the analysis, resulting in a selection of 103 bands for further examination. The dataset encompasses a spatial domain of 610 $\times$ 340 pixels, offering a spatial resolution of 1.3 m. It encompasses a categorial representation of nine distinct classes. The false color image and the classification groundtruth are shown in Fig. 7.

The WHHH dataset [57], [58] was acquired on November 20, 2017, from 16:23 to 17:37 in Honghu City, Hubei province, China, using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform. The UAV operated at an altitude of 100 m, capturing images with a size of 940 $\times$ 475 pixels. A total of 270 spectral bands (ranging from 400 to 1000 nm) were used in the dataset, providing a spatial resolution of approximately 0.043 m for the UAV-borne hyperspectral imagery. The captured images represent a complex agricultural scene, featuring various types of crops. The region includes different varieties of the same crop, such as Chinese cabbage and cabbage, as well as Brassica chinensis and small Brassica chinensis. In total, the dataset comprises 22 different classes of land cover. The false color image and the classification groundtruth are shown in Fig. 8.

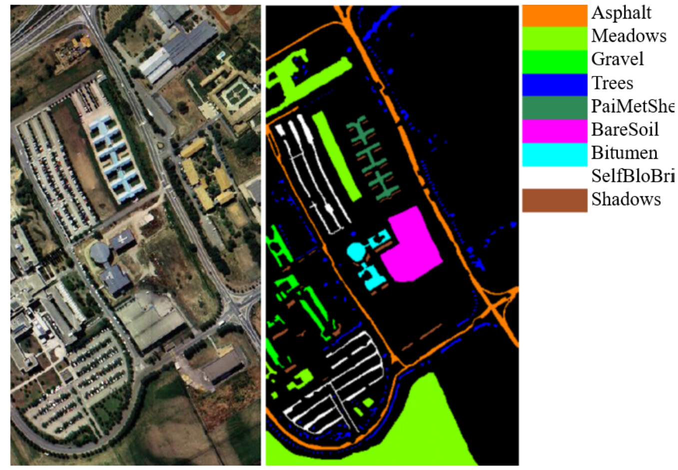

Fig. 7. False-color composite image, the corresponding ground-truth map, and the legend of the PU dataset.

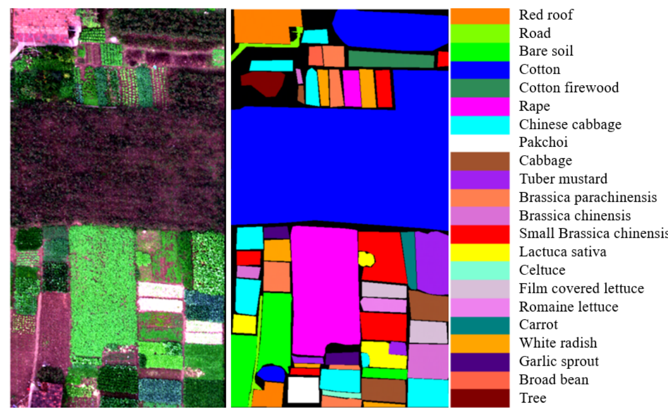

Fig. 8. False-color composite image, the corresponding ground-truth map, and the legend of the WHHH dataset.

Table I shows the size of the training and testing datasets used in this experiment, including the sample distribution of each land cover category.

2) Evaluation Metrics: We will compare the effectiveness of our algorithm with other algorithms from four aspects, mainly including overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and the classification accuracy of each land cover category itself.

3) Training Details: The PyTorch training framework was utilized for the implementation and training of the model, with the basic hardware environment comprising an Intel[1], Xeon[1], Platinum 8336C CPU at 2.30 GHz, 64 cores, 128 GB

[1]Registered trademark.

TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE IP DATASET, THE PU DATASET, AND THE WHHH DATASET

No.	Indian Pines			Pavia University			WHU-Hi-HongHu dataset		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	10	36	Asphalt	10	6621	Red roof	10	14031
2	CornMotill	10	1418	Meadows	10	18639	Road	10	3502
3	CornMintill	10	820	Gravel	10	2089	Bare soil	10	21811
4	Corn	10	227	Trees	10	3054	Cotton	10	163275
5	GrassPasture	10	473	PaintedMetalSheets	10	1335	Cotton firewood	10	6208
6	GrassTrees	10	720	BareSoil	10	5019	Rape	10	44547
7	GrassPastureMowed	10	18	Bitumen	10	1320	Chinese cabbage	10	24093
8	HayWindrowed	10	468	SelfBlockingBricks	10	3672	Pakchoi	10	4044
9	Oats	10	10	Shadows	10	937	Cabbage	10	10809
10	SoybeanNotill	10	962				Tuber mustard	10	12384
11	SoybeanMintill	10	2445				Brassica parachinensis	10	11005
12	SoybeanClean	10	583				Brassica chinensis	10	8944
13	Wheat	10	195				Small Brassica chinensis	10	22497
14	Woods	10	1255				Lactuca sativa	10	7346
15	BuildingsGrassTreesDrives	10	376				Celtuce	10	992
16	StoneSteelTowers	10	83				Film covered lettuce	10	7252
17							Romaine lettuce	10	3000
18							Carrot	10	3207
19							White radish	10	8702
20							Garlic sprout	10	3476
21							Broad bean	10	1318
22							Tree	10	4030
	Total	160	10089	Total	90	42686	Total	220	386473

of memory, and two NVIDIA GeForce RTX 4090 GPUs, each with 24 GB of memory.

We employ the Adam optimizer with a learning rate of 1e-3, which effectively guides the learning process. The batch size, a crucial factor affecting both memory consumption and training efficiency, is dynamically altered based on the available GPU memory. To ensure optimal processing, we assign batch sizes of 64, 64, and 32, respectively, for the IP, PU, and WHHH datasets, aligning with their unique characteristics. Moreover, within our Transformer architecture, the embedding dimension is set to 64, allowing for an effective representation of the input data. Additionally, Section III-A4 delves into comprehensive discussions on significant parameters, including the patch size, the number of transformer blocks and the number of channels preserved by PCA.

4) Parameter Analysis: We conducted a detailed analysis of important parameters, including the number of blocks used in the Transformer, the patch size of the input data, and the number of channels preserved by PCA. Fig. 9 illustrates the experimental results of the search for optimal parameter settings. We found that different datasets may have different optimal parameter choices. For the number of transformer blocks, the optimal parameters for the IP, PU, and WHHH datasets were found to be 3, 3, and 2, respectively. Furthermore, we observed that the impact of this parameter on the IP dataset was lower compared to its impact on the PU and WHHH datasets. Regarding the patch size parameter, our algorithm tends to perform better with relatively larger patch sizes. From the experiment, it can be observed that the OA is relatively lower for all three datasets when using smaller patch sizes such as 5×5 and 9×9. As the patch size increases, the OA change is not significant for the IP dataset, increases initially and then slowly decreases for the PU dataset, and continuously increases for the WHHH dataset. The optimal patch sizes for the IP, PU, and WHHH datasets were found to be 13, 13, and 21, respectively. The number of channels preserved by PCA also had an impact on all three datasets. Additionally, it was found through experimentation that the optimal number of PCA channels for the IP, PU, and WHHH datasets were 200, 10, and 30, respectively.

B. Performance Analysis

To evaluate the effectiveness of the proposed method, a variety of representative algorithms were selected as comparative experiments, including spectral-spatial residual network (SSRN) [59], SpectralFormer [49], HiT [51], SS1DSwin [52], SSFTT [53], CASST [54], and LSGA [55].

1) SSRN [59] enhances the effectiveness of CNN-based models by utilizing the SSRN to fuse and extract spectral-spatial information. SSRN represents one of the state-of-the-art CNN-based methods, and it was chosen as a representative algorithm of the CNN class for comparison in this article.
2) SpectralFormer [49] is one of the most popular transformer-based algorithms and is also one of the most commonly used comparative algorithms in recent years.
3) HiT [51], based on the ViT approach, processes spectral-spatial information by embedding convolution operations into the transformer and represents a representative transformer-based algorithm for HSI classification tasks.

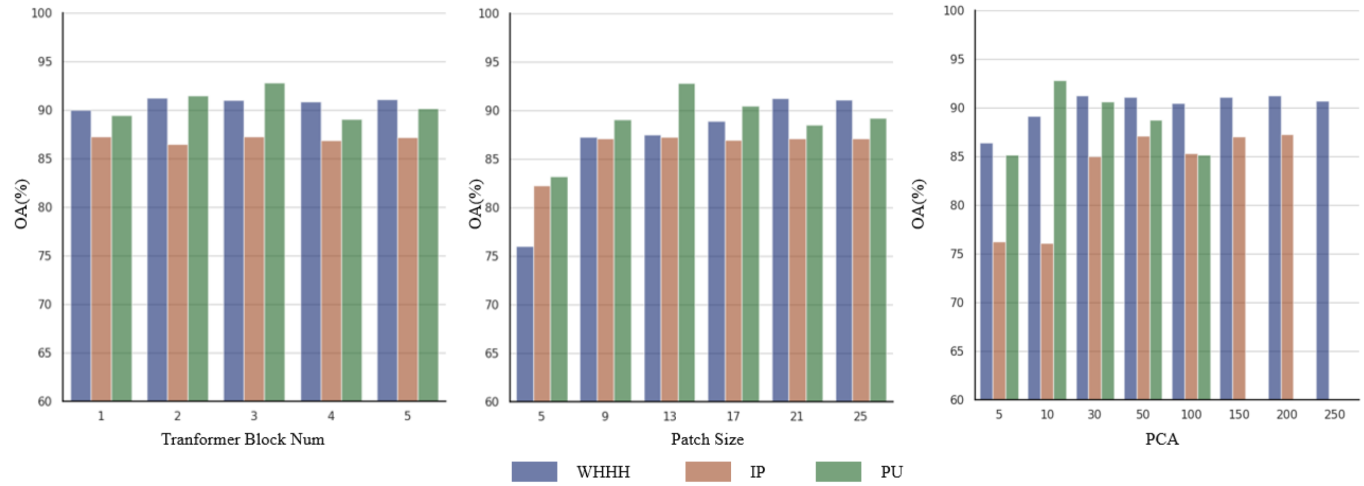

Fig. 9. Impact of select hyperparameters on OA of three datasets (IP, PU, WHHH). (a) Impact of transformer block num on OA Performance. (b) Impact of patch size on OA Performance. (c) Impact of PCA num on OA performance.

TABLE II
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE IP DATASET (TEN LABELS PER-CLASS)

Class	SSRN (2018)	SpectralFormer (2022)	HiT (2022)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	Ours
1	**100.00 ± 0.00**	87.22 ± 11.21	99.44 ± 1.24	88.89 ± 5.20	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.44 ± 1.24
2	70.89 ± 12.57	24.30 ± 4.44	40.42 ± 11.73	38.60 ± 4.32	75.98 ± 2.67	**80.37 ± 2.58**	49.37 ± 3.59	78.87 ± 2.06
3	77.95 ± 3.27	52.07 ± 5.52	31.17 ± 6.67	62.44 ± 12.55	**84.63 ± 1.58**	80.15 ± 3.19	84.17 ± 1.81	82.56 ± 1.26
4	97.71 ± 1.75	55.77 ± 13.32	63.35 ± 24.62	61.06 ± 11.50	93.30 ± 3.83	89.07 ± 2.30	**99.91 ± 0.20**	99.21 ± 0.65
5	93.78 ± 3.05	50.15 ± 15.23	38.09 ± 4.69	79.20 ± 10.62	94.71 ± 1.11	92.56 ± 1.13	84.14 ± 1.90	**96.62 ± 1.46**
6	94.08 ± 1.66	54.92 ± 3.79	73.22 ± 9.47	63.33 ± 29.82	93.36 ± 0.58	**96.50 ± 1.96**	95.83 ± 1.47	94.31 ± 0.86
7	**100.00 ± 0.00**	87.78 ± 15.42	**100.00 ± 0.00**	88.89 ± 8.78	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	99.96 ± 0.10	82.39 ± 8.86	98.58 ± 1.31	92.69 ± 2.03	**100.00 ± 0.00**	**100.00 ± 0.00**	99.91 ± 0.19	**100.00 ± 0.00**
9	**100.00 ± 0.00**	**100.00 ± 0.00**	98.00 ± 4.47	96.00 ± 8.94	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
10	86.51 ± 2.42	52.93 ± 9.01	60.09 ± 16.08	56.11 ± 4.21	79.83 ± 3.65	75.24 ± 1.48	79.50 ± 2.66	**87.61 ± 2.10**
11	68.76 ± 5.47	32.25 ± 10.66	56.92 ± 9.22	44.79 ± 7.99	67.46 ± 4.00	65.43 ± 1.36	69.22 ± 3.73	**75.43 ± 3.27**
12	87.48 ± 2.20	26.52 ± 6.36	32.71 ± 10.14	51.70 ± 7.91	81.37 ± 6.01	81.17 ± 5.04	81.06 ± 5.54	**93.17 ± 0.89**
13	99.28 ± 0.78	74.36 ± 21.55	81.54 ± 5.54	97.13 ± 0.58	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	98.15 ± 0.99	62.71 ± 22.16	84.01 ± 16.40	83.38 ± 5.89	97.91 ± 0.51	95.38 ± 2.29	89.96 ± 2.24	**98.73 ± 0.16**
15	96.06 ± 3.01	38.56 ± 10.61	44.30 ± 8.24	78.14 ± 12.30	98.35 ± 0.55	96.86 ± 1.57	85.21 ± 4.42	**99.15 ± 0.61**
16	98.80 ± 2.09	97.59 ± 2.95	93.49 ± 4.48	98.55 ± 1.98	**100.00 ± 0.00**	99.28 ± 1.62	**100.00 ± 0.00**	**100.00 ± 0.00**
OA(%)	83.37 ± 1.88	45.44 ± 3.35	58.79 ± 4.74	60.14 ± 5.04	83.30 ± 1.35	82.26 ± 0.59	78.24 ± 0.92	**87.31 ± 0.83**
AA(%)	91.84 ± 0.78	61.22 ± 1.85	68.46 ± 3.02	73.81 ± 3.36	91.68 ± 0.63	90.75 ± 0.57	88.64 ± 0.60	**94.07 ± 0.29**
$\kappa \times 100$	81.31 ± 2.07	39.90 ± 3.41	53.24 ± 5.16	55.64 ± 5.48	81.20 ± 1.49	80.04 ± 0.65	75.41 ± 1.00	**85.67 ± 0.92**

4) SS1DSwin [52] mainly consists of a group-wise feature tokenization module (GFTM) and a 1DSwin transformer with cross-block normalized connection module (TCNCM). It is one of the latest representative transformer-based algorithms for HSI classification.
5) SSFTT [53] effectively combines CNN and transformer and introduces a Gaussian-weighted feature tokenizer to further enhance the results. It is one of the transformer-based algorithms with Competitive effectiveness in the current HSI classification task.
6) CASST [54] effectively integrates spectral-spatial information through the Cross-Attention module and is one of the transformer-based algorithms with Competitive effectiveness in the current HSI classification task.
7) LSGA [55] is one of the latest transformer-based algorithms that is a ViT based on the LSGA mechanism, which extracts global deep semantic features and has good performance in HSI classification tasks.

1) Quantitative Results: Tables II–IV present the classification performance of the IP, PU, and WHHH datasets, including OA, AA, κ, and accuracy for each class. The best performances are highlighted in bold.

It can be observed that our proposed model exhibits the best overall performance on all three datasets. Under the condition of using ten samples per class as the training set, the SpectralFormer, HiT, and SS1DSwin models show lower performance compared to other algorithms, indicating room for further improvement when using limited training samples. The SSRN, SSFTT, CASST, and LSGA algorithms demonstrate higher classification accuracy, suggesting their strong classification capabilities in limited sample training set scenarios. Our method further enhances the classification performance, with relative improvements of 3.94%, 1.35%, and 3.67% in OA compared to the second-best method on the IP, PU, and WHHH datasets, respectively.

Specifically, for different subclasses of the same dataset, our proposed model still outperforms the comparative models in the majority of categories. Further analysis was conducted for the subclasses where our model showed relatively biased performance compared to the comparative model. For the IP and WHHH datasets, we found that in the subclasses where

TABLE III

CLASSIFICATION RESULTS OF EXPERIMENTS ON THE PU DATASET (TEN LABELS PER-CLASS)

Class	SSRN (2018)	SpectralFormer (2022)	HiT (2022)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	Ours
1	75.80 ± 30.18	38.18 ± 5.61	58.06 ± 7.64	64.36 ± 8.47	71.89 ± 4.95	85.51 ± 2.00	81.52 ± 0.86	**90.83 ± 2.23**
2	72.97 ± 23.22	75.18 ± 1.96	85.53 ± 6.78	72.44 ± 3.73	89.71 ± 1.87	91.91 ± 1.15	94.15 ± 1.81	**95.17 ± 0.79**
3	91.89 ± 16.25	60.58 ± 1.33	42.72 ± 15.02	57.78 ± 11.16	90.33 ± 1.49	**92.99 ± 4.33**	88.84 ± 3.47	66.64 ± 3.69
4	91.89 ± 5.21	87.25 ± 4.29	92.52 ± 3.06	85.80 ± 1.90	**93.46 ± 2.60**	83.24 ± 4.78	92.55 ± 2.41	91.51 ± 1.65
5	98.20 ± 2.50	99.94 ± 0.03	99.97 ± 0.04	99.99 ± 0.03	99.19 ± 1.28	98.58 ± 1.17	**100.00 ± 0.00**	99.93 ± 0.09
6	**99.65 ± 0.79**	54.18 ± 7.07	60.62 ± 14.84	63.34 ± 8.30	97.48 ± 1.85	97.68 ± 2.19	96.17 ± 2.70	97.27 ± 0.85
7	95.14 ± 3.39	91.61 ± 1.56	89.92 ± 6.04	75.05 ± 3.74	98.24 ± 1.24	**99.95 ± 0.10**	99.92 ± 0.13	99.14 ± 1.89
8	81.54 ± 14.94	60.73 ± 1.22	65.16 ± 13.23	73.17 ± 7.54	59.26 ± 6.50	53.15 ± 5.37	82.60 ± 7.36	**88.14 ± 2.85**
9	90.25 ± 6.19	97.31 ± 1.48	95.45 ± 3.65	88.30 ± 3.86	97.99 ± 1.72	97.50 ± 1.21	97.74 ± 0.47	**99.94 ± 0.06**
OA(%)	81.42 ± 10.09	67.65 ± 1.05	75.52 ± 5.10	71.71 ± 1.39	86.28 ± 1.61	88.27 ± 0.71	91.50 ± 0.68	**92.85 ± 0.44**
AA(%)	88.59 ± 2.58	73.89 ± 0.26	76.66 ± 3.95	75.58 ± 1.49	88.62 ± 1.32	88.95 ± 0.68	**92.61 ± 0.35**	92.06 ± 0.58
$\kappa \times 100$	77.01 ± 11.56	58.72 ± 1.10	68.75 ± 6.21	63.82 ± 1.64	82.24 ± 2.04	84.71 ± 0.89	88.87 ± 0.86	**90.61 ± 0.57**

TABLE IV

CLASSIFICATION RESULTS OF EXPERIMENTS ON THE WHHH DATASET (TEN LABELS PER-CLASS)

Class	SSRN (2018)	SpectralFormer (2022)	HiT (2022)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	Ours
1	**94.71 ± 1.68**	77.43 ± 0.89	93.93 ± 1.16	88.89 ± 3.13	91.93 ± 1.04	93.03 ± 0.49	93.88 ± 1.69	88.99 ± 2.36
2	79.26 ± 7.25	52.20 ± 6.45	73.79 ± 6.87	82.07 ± 1.31	81.52 ± 4.43	78.66 ± 2.88	87.11 ± 2.51	**95.55 ± 1.28**
3	**83.96 ± 1.92**	78.44 ± 3.08	78.99 ± 3.86	72.94 ± 1.71	80.72 ± 0.68	82.57 ± 0.45	82.16 ± 2.03	81.53 ± 3.20
4	97.71 ± 1.46	69.03 ± 7.81	92.16 ± 0.75	84.48 ± 6.84	90.73 ± 1.06	93.12 ± 0.78	90.59 ± 2.21	**98.00 ± 1.16**
5	90.68 ± 3.82	67.44 ± 0.62	86.60 ± 2.55	70.61 ± 10.30	92.68 ± 2.32	93.12 ± 1.39	92.28 ± 2.06	**94.57 ± 1.67**
6	89.36 ± 3.51	65.68 ± 1.30	83.80 ± 2.79	79.27 ± 4.31	91.84 ± 0.74	91.96 ± 0.76	91.84 ± 0.72	**97.25 ± 1.13**
7	63.90 ± 15.99	35.27 ± 1.68	50.72 ± 3.38	52.45 ± 7.26	66.44 ± 2.71	68.16 ± 4.33	68.28 ± 2.75	**73.62 ± 2.52**
8	77.50 ± 7.51	23.29 ± 2.17	33.90 ± 15.23	35.30 ± 2.90	65.44 ± 3.07	69.02 ± 2.02	84.07 ± 1.56	**90.45 ± 1.44**
9	93.59 ± 1.08	88.74 ± 1.70	82.86 ± 8.08	88.12 ± 7.64	93.85 ± 0.87	**94.79 ± 0.08**	94.13 ± 0.29	94.07 ± 0.75
10	80.30 ± 6.94	37.48 ± 10.71	33.34 ± 9.86	53.81 ± 10.71	70.90 ± 7.99	78.94 ± 1.00	75.44 ± 3.64	**84.95 ± 3.35**
11	51.38 ± 18.29	28.04 ± 2.21	23.10 ± 5.77	46.22 ± 1.29	61.42 ± 3.28	66.52 ± 1.32	**70.13 ± 5.55**	67.93 ± 3.34
12	67.82 ± 20.81	56.88 ± 5.24	60.77 ± 5.87	54.16 ± 3.00	58.37 ± 5.38	70.54 ± 1.82	64.43 ± 3.52	**88.23 ± 1.94**
13	55.45 ± 28.57	34.53 ± 4.64	47.01 ± 5.83	48.43 ± 4.08	48.20 ± 7.27	43.28 ± 3.87	61.22 ± 4.73	**69.80 ± 6.52**
14	92.90 ± 5.64	57.56 ± 2.18	60.74 ± 5.73	70.60 ± 9.26	92.56 ± 2.38	96.20 ± 0.63	68.95 ± 1.08	**96.57 ± 1.40**
15	**99.42 ± 0.84**	74.54 ± 3.70	92.24 ± 2.56	86.36 ± 5.32	98.13 ± 1.11	98.19 ± 0.61	96.83 ± 1.07	97.32 ± 0.20
16	87.37 ± 27.02	57.76 ± 9.71	27.08 ± 9.00	68.34 ± 5.34	90.44 ± 1.62	84.22 ± 1.64	**97.23 ± 1.89**	89.20 ± 4.81
17	99.56 ± 0.39	71.41 ± 3.52	74.27 ± 10.87	74.02 ± 3.22	96.87 ± 0.85	96.54 ± 0.60	95.03 ± 0.64	**99.83 ± 0.23**
18	**99.42 ± 0.56**	33.93 ± 18.76	63.04 ± 10.48	70.23 ± 2.80	99.40 ± 0.34	99.19 ± 0.30	98.30 ± 0.50	96.99 ± 1.13
19	86.73 ± 10.97	67.34 ± 5.83	57.44 ± 7.63	78.81 ± 2.47	89.43 ± 1.84	**90.67 ± 2.54**	85.73 ± 4.47	82.51 ± 2.01
20	**98.04 ± 0.97**	77.72 ± 4.09	74.39 ± 1.74	80.84 ± 11.67	83.14 ± 5.55	90.34 ± 1.32	96.67 ± 0.53	**98.04 ± 1.11**
21	98.82 ± 1.14	49.88 ± 4.17	63.57 ± 10.32	87.08 ± 2.01	91.53 ± 2.88	93.12 ± 3.38	94.83 ± 1.61	**98.98 ± 0.97**
22	90.67 ± 6.39	53.72 ± 3.84	86.23 ± 4.55	78.00 ± 4.42	86.53 ± 5.03	88.32 ± 0.84	95.23 ± 1.80	**99.96 ± 0.04**
OA(%)	87.55 ± 4.48	61.93 ± 4.44	76.31 ± 1.34	74.88 ± 4.45	83.93 ± 0.36	85.71 ± 0.43	85.47 ± 1.64	**91.22 ± 0.50**
AA(%)	85.39 ± 5.90	57.20 ± 1.29	65.45 ± 1.75	70.50 ± 2.80	82.82 ± 0.68	84.57 ± 0.39	85.65 ± 1.07	**90.20 ± 0.35**
$\kappa \times 100$	84.35 ± 5.68	55.18 ± 4.43	70.65 ± 1.66	69.45 ± 5.01	80.10 ± 0.42	82.24 ± 0.52	82.02 ± 1.95	**88.95 ± 0.61**

our algorithm performed relatively worse, the difference in performance compared to the highest-performing classification metric in that subclass was relatively small. However, on the PU dataset, our algorithm exhibited a significant gap compared to the best-performing algorithm, CASST, in the Gravel category (CASST: 91.99%, Ours: 66.64%). Through analysis of the confusion matrix, we discovered that our model incorrectly classified some parts of Gravel as Bricks. We believe that the model proposed in this article has further enhanced spectral information, thus possibly leading to misclassification in cases where there is a similarity in spectral information but stronger reliance on pure spatial information.

We also validated the model's performance under different training sample sizes. As shown in Fig. 10, we adjusted the sample sizes used for training each land cover class to assess the classification performance of different algorithms. The experiment showed that as the sample size increased, the classification accuracy of each model improved to varying degrees. Notably, the model proposed in this article demonstrated competitive performance across different training sample sizes for the IP, PU, and WHHH datasets. This experiment further confirms the effectiveness of the proposed method.

2) Visual Evaluation: Figs. 11–13, respectively, show the final classification results of the IP, PU, and WHHH datasets. Our proposed model exhibits minimal noise points and the best classification results in terms of visual effects. Specifically, the SpectralFormer, HiT, and SS1DSwin models have more misclassification results, with instances of misclassifying similar land covers within the same cluster. In contrast, the SSRN, SSFTT, CASST, and LSGA methods show significant improvement, reducing the occurrence of noise points in the classification result. Our model further enhances the classification performance based on these methods, particularly in scenarios with a high density of land covers within a unit space (e.g., upper-left part of IP, surroundings of PU Bricks, lower-right part of WHHH). Our model exhibits superior classification accuracy compared to other methods in such scenarios.

C. Ablation Study

To further validate the effectiveness of the RIPE and SSCA modules, an ablation study were conducted. In these experiments, while modifying the corresponding modules,

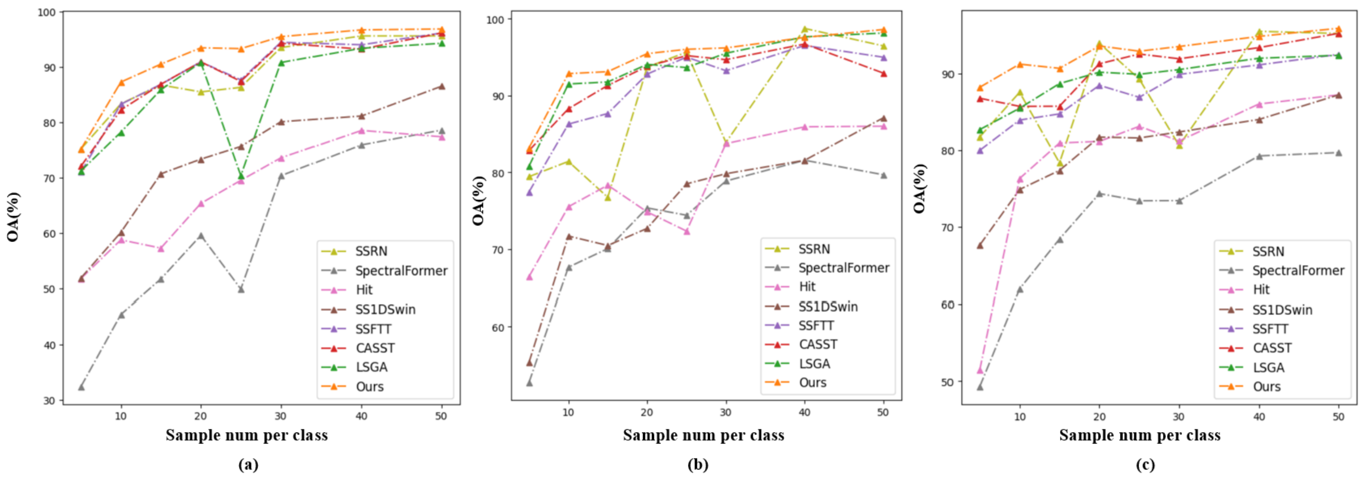

Fig. 10. Overall classification accuracy of various algorithms under different training sample sizes. (a) Indian Pines. (b) Pavia University. (c) WHU-Hi-HongHu.

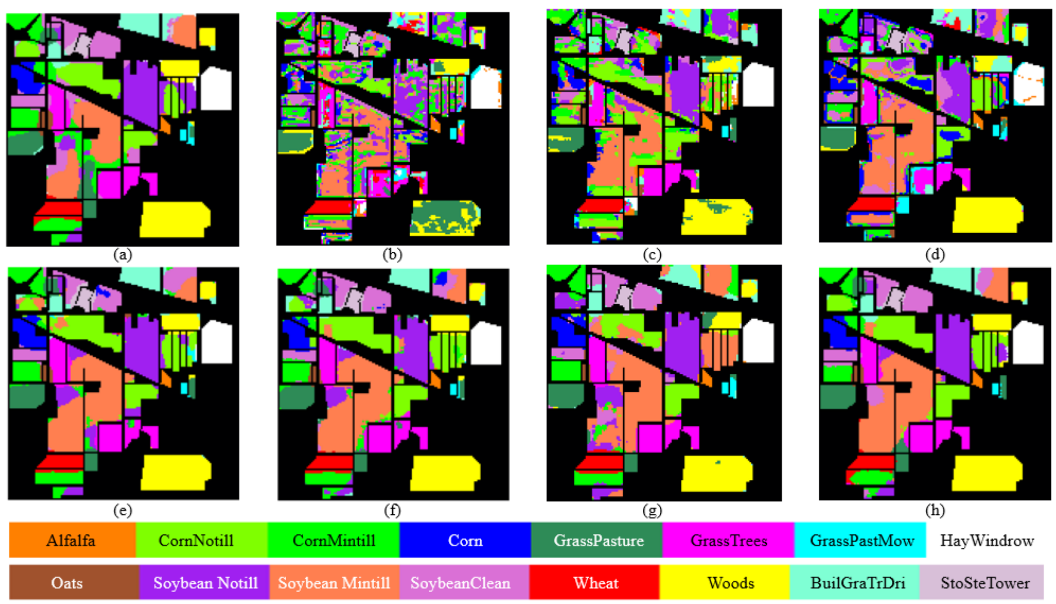

Fig. 11. Visual classification results of IP with compared methods. (a) SSRN. (b) SpectralFormer. (c) HiT. (d) SS1DSwin. (e) SSFTT. (f) CASST. (g) LSGA. (h) Ours.

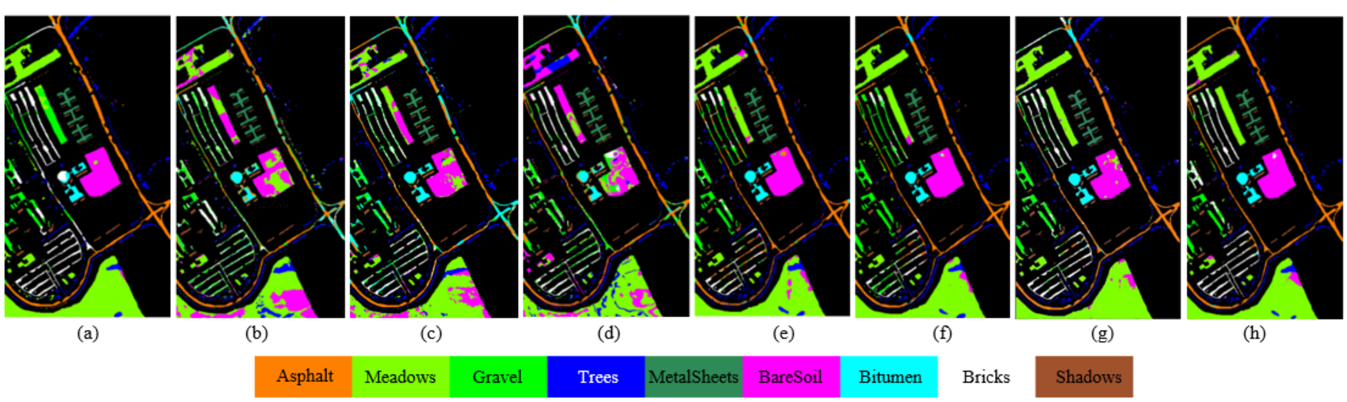

Fig. 12. Visual classification results of PU from compared methods. (a) SSRN. (b) SpectralFormer. (c) HiT. (d) SS1DSwin. (e) SSFTT. (f) CASST. (g) LSGA. (h) Ours.

other aspects such as network framework, hyperparameters, and dataset division remained consistent. The proposed RIPE module and SSCA module in this article each consist of two sub-structures. The RIPE module includes RR and center relative PE. The SSCA module includes CA Transformer and Fusion (FU). To conduct detailed effect validation of

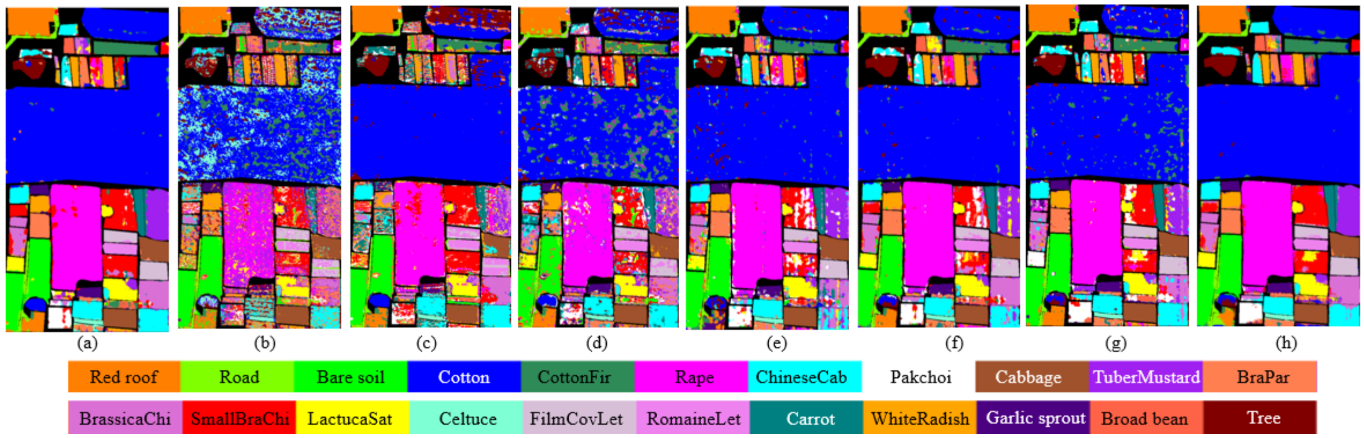

Fig. 13. Visual classification results of WHHH from compared methods. (a) SSRN. (b) SpectralFormer. (c) HiT. (d) SS1DSwin. (e) SSFTT. (f) CASST. (g) LSGA. (h) Ours.

TABLE V

CLASSIFICATION PERFORMANCE OF THE PROPOSED CASST METHOD WITH DIFFERENT ABLATION STRATEGIES ON THE THREE HSI DATASETS

Index	RIPE		SSCA		Indian Pines			Pavia University			WHU-HI-HONGHU		
	RR	PE	CA	FU	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
1	✓	✓	✓	✓	87.31 ± 0.83	94.07 ± 0.29	85.67 ± 0.92	92.85 ± 0.44	92.06 ± 0.58	90.61 ± 0.57	91.22 ± 0.50	90.20 ± 0.35	88.95 ± 0.61
2	×	×	×	×	85.21 ± 0.25	92.66 ± 0.19	83.35 ± 0.27	88.11 ± 2.29	90.02 ± 1.34	84.57 ± 2.84	88.55 ± 0.48	86.86 ± 0.51	85.66 ± 0.57
3	✓	✓	✓	×	86.22 ± 0.82	93.61 ± 0.27	84.47 ± 0.90	90.90 ± 1.57	91.36 ± 0.71	88.04 ± 1.97	91.15 ± 0.30	89.86 ± 0.34	88.86 ± 0.36
4	✓	✓	×	×	85.75 ± 0.39	92.80 ± 0.30	83.95 ± 0.42	88.22 ± 3.17	88.87 ± 1.67	84.68 ± 3.86	90.98 ± 0.47	89.52 ± 0.54	88.65 ± 0.58
5	×	✓	✓	✓	87.11 ± 0.91	94.15 ± 0.41	85.47 ± 1.01	90.07 ± 1.17	90.47 ± 1.19	86.89 ± 1.49	89.19 ± 0.57	86.28 ± 0.79	86.38 ± 0.73
6	✓	×	✓	✓	86.73 ± 0.66	93.91 ± 0.42	85.04 ± 0.73	90.14 ± 0.99	90.85 ± 0.70	87.01 ± 1.24	91.01 ± 0.79	89.52 ± 0.98	88.66 ± 1.00
7	×	×	✓	✓	86.59 ± 0.29	93.66 ± 0.33	84.89 ± 0.32	90.22 ± 1.41	90.73 ± 0.47	87.11 ± 1.76	89.26 ± 0.70	86.51 ± 1.24	86.44 ± 0.89

· RR: random central rotation, PE: center relative position embedding, CA: Center-Aware Transformer, FU: Fusion

the involved sub-structures in the article, we designed seven scenarios, specifically including as follows.

1) Our proposed model.
2) Removing the RIPE and SSCA modules (i.e., excluding the four sub-structures: RR, PE, CA, and FU).
3) Only removing the FU sub-structure.
4) Removing the SSCA module, including simultaneously removing the CA and FU sub-structures (note that the FU sub-structure is dependent on the CA structure, thus it becomes inactive when the CA structure is removed).
5) Only removing the RR sub-structure.
6) Only removing the PE sub-structure.
7) Removing the RIPE module, including simultaneously removing the RR and PE sub-structures.

Table V presents the results of ablation experiments. Firstly, we conducted a comprehensive analysis of the RIPE and SSCA modules. Observing experiments 1, 2, 4, and 7, it can be noted that removing both the RIPE and SSCA modules resulted in performance losses of 2.10%, 4.74%, and 2.67% on the IP, PU, and WHHH datasets, respectively. Specifically, when only the SSCA module was removed, performance losses of 1.56%, 4.63%, and 0.24% were observed on the IP, PU, and WHHH datasets, respectively. Conversely, removing only the RIPE module led to performance losses of 0.72%, 2.63%, and 1.96% on the IP, PU, and WHHH datasets, respectively. The data indicates that the SSCA module has a stronger impact on the model compared to the RIPE module for the IP and PU datasets, while the WHHH dataset exhibits the opposite trend.

Through Table V, we can further analyze the impact of sub-structures within the RIPE and SSCA modules. From experiments 1, 3, and 4, we observed that removing only the FU structure resulted in performance losses of 1.09%, 1.95%, and 0.07% on the IP, PU, and WHHH datasets, respectively. Taking experiment 4 as the base, we can indirectly estimate that the CA structure approximately contributed performance gains of 0.47%, 2.68%, and 0.17% on the IP, PU, and WHHH datasets, respectively, as observed in experiment 5. Comparing the performance losses between experiments 1 and 4, we believe that the impact of the Fusion structure on the IP and PU datasets is comparable to that of the CA Transformer, while its effect on WHHH is weaker. This discrepancy may be attributed to the relatively small number of Transformer blocks in WHHH (only two layers).

Using experiments 1, 5, 6, and 7, we further analyzed the impact of individual sub-structures within the RIPE module. Specifically, on the IP, PU, and WHHH datasets, removing only the RR structure resulted in performance losses of 0.20%, 2.78%, and 2.03%, respectively. Similarly, removing only the PE structure led to performance losses of 0.58%, 2.71%, and 0.21% on the IP, PU, and WHHH datasets, respectively. We observed that the RR and PE structures have similar effects on the IP and PU datasets, while the PE structure has only a weak influence on WHHH. Conversely, the RR structure significantly impacts the classification performance on the WHHH dataset.

As shown in Fig. 14, we take the IP dataset as an example to verify the OA loss variation caused by removing the RIPE and

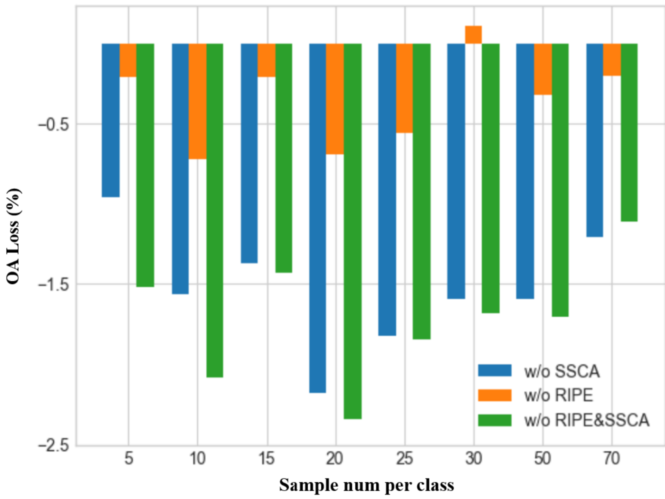

Fig. 14. OA loss of different ablation strategies compared to our proposed model on the IP dataset at different sample sizes.

SSCA modules under different training sample sizes. From the figure, it can be observed that, under different training sample sizes, the impact of the SSCA module on the model in the IP dataset consistently exceeds that of the RIPE module. Furthermore, as the sample size continues to increase, the loss caused by removing the SSCA module initially fluctuates and then gradually decreases. It is also notable that, with an increase in sample size, the loss caused by removing the RIPE module tends to decrease continuously.

IV. Conclusion

In the current landscape of HSI classification research, two primary challenges persist in spectral-spatial feature extraction: 1) interference caused by absolute PE and 2) disruption of central spectral vectors due to the introduction of irrelevant spatial information. This article proposed a novel framework, the SQSFormer, to address these challenges. The framework comprises the RIPE module and the SSCA module. The RIPE module achieves decoupling of spectral-spatial features from absolute PE by leveraging center relative PE and random center rotation. This effectively resolves the issue of introducing unnecessary spatial noise through absolute PE. The SSCA module, on the other hand, addresses interference caused by irrelevant spatial information by integrating spectral-spatial features from multiple layers of CA transformers. This facilitates multiscale spectral-spatial feature extraction while minimizing the introduction of irrelevant spatial information.

In future research, given the continuous emergence of new land features in actual remote sensing scenarios, we will explore how to align spectral-spatial features with semantic features corresponding to class labels based on SQSFormer. The ultimate goal is to achieve zero-shot classification in hyperspectral imagery.

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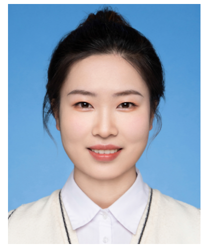

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